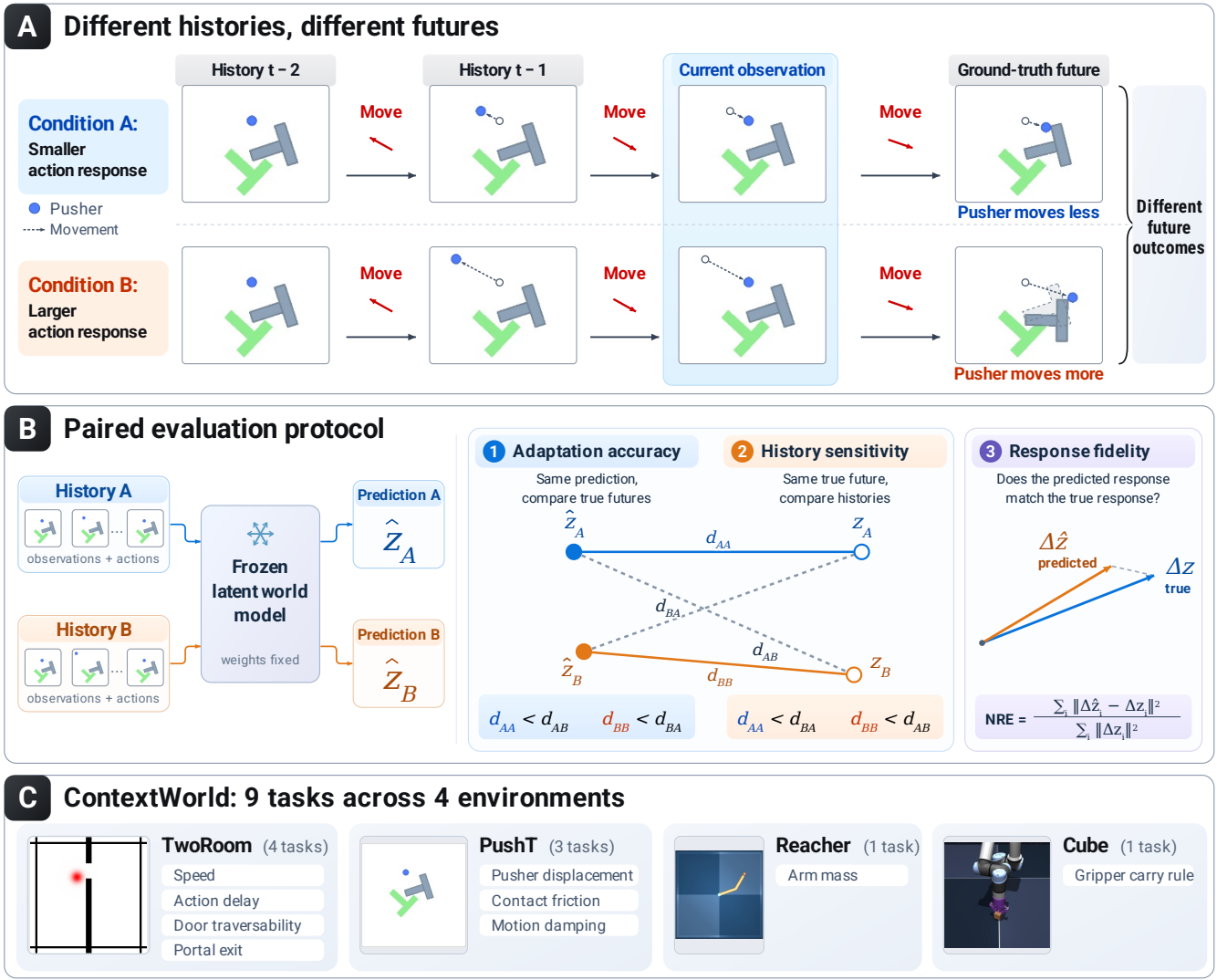

A
Different histories, different futures
History t − 2
History t − 1
Current observation
Ground-truth future
Condition A:
Smaller
action response
Pusher
Movement
Move
Move
Move
Pusher moves less
Condition B:
Larger
action response
Move
Move
Move
Pusher moves more
Different
future
outcomes
B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Frozen
latent world
model
weights fixed
Prediction A
$\hat{z}_A$
Prediction B
$\hat{z}_B$
1 Adaptation accuracy
Same prediction,
compare true futures
2 History sensitivity
Same true future,
compare histories
$\hat{z}_A$
z_A
d_{AA}
d_{BA}
d_{AB}
d_{BB}
$\hat{z}_B$
z_B
$d_{AA} < d_{AB}$
$d_{BB} < d_{BA}$
$d_{AA} < d_{BA}$
$d_{BB} < d_{AB}$
3 Response fidelity
Does the predicted response
match the true response?
$\Delta\hat{z}$
predicted
Δz
true
NRE = $\frac{\sum_i \|\Delta\hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}$
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule